

VROOM System Architecture & Capabilities Report

From Routing MVP to Persistent Dispatch Orchestrator

4

Architecture Pillars

2,500+

Max Matrix Size

Docker

Deployment Model

1 Executive Summary

The VROOM platform has undergone a strategic architectural evolution, transitioning from a reactive, session-based routing utility into a **proactive, AI-enhanced dispatch orchestrator**. This shift moves the system beyond solving isolated Vehicle Routing Problems (VRP) toward managing a persistent, stateful workforce environment. By integrating temporal awareness and high-scale local execution, the platform provides enterprises with a blueprint for dynamic logistics management that prioritises operational continuity over ephemeral computation.

Next-Gen AI Skill Parsing	Architectural Data Persistence	Temporal Traffic Integration
Automated translation of natural language competencies into structured solver constraints.	Transition from one-off file parsing to a stateful model for Jobs and Engineers, enabling multi-day planning horizons.	Real-time synchronisation with dynamic speed modifiers to ensure route consistency in volatile urban environments.

2 Core Platform Architecture: The Four Pillars of Orchestration

The transition to a modular, high-scale architecture addresses the inherent limitations of legacy routing tools by replacing temporary parsing logic with a persistent **"data bridge"**. This foundation allows the orchestrator to maintain the state of the field service environment, ensuring that re-optimisation cycles are grounded in verified historical and real-time data.

2.1 Persistent Data Integration Bridge

The system has moved away from legacy one-off file parsing toward a persistent management workflow for **Jobs** (tasks) and **Engineers** (vehicle rosters). A stateful database maintains long-term records of technician availability and task backlogs.

Architectural Insight: The bridge relies on the `location_index` property. When using custom matrices, `location_index` is the mandatory pointer for duration and cost lookups, while the `location` coordinates array remains optional for response-layer mapping. This enables reliable state across multi-day planning horizons.

2.2 AI-Driven Skill Classification Engine

The platform integrates an Anthropic-based Next-Gen AI engine that parses complex job requirements and engineer certifications expressed in natural language — for example, *"Level 3 HVAC Tech"* — and maps them dynamically to the skills arrays required by the VROOM solver. The engine performs semantic validation and contextual mapping, automatically scaling as new competencies are introduced without any manual maintenance overhead.

2.3 Temporal Traffic Engine & Routing Logic

The platform integrates **TomTom-driven dynamic traffic modifiers**, moving beyond static distance-based routing. Precise Unix Timestamps and the ratio `currentSpeed / freeFlowSpeed` enable realistic ETAs. This logic is implemented via Valhalla's `speed_types` (`freeflow`, `constrained`, `predicted`, `current`), allowing the solver to proactively adjust for time-dependent congestion.

2.4 Local Dockerised Solver & High-Scale Execution

A containerised vroom-express environment eliminates external API latency and cost. Python-based stress tests verify performance for matrices exceeding **2,500 locations**. The backend configures OSRM with the `--max-table-size` flag, bypassing default routing engine limits so the local solver can compute massive duration matrices without the overhead or throttling of third-party SaaS providers.

3 Advanced Logic & Constraint Framework

Field service logistics demand a sophisticated logic framework to manage the complexities of human workforces and multi-day temporal horizons.

Multi-Day Shift Planning Utilises the `time_windows` array within job objects and the `time_window` object for vehicles to manage complex shifts, defining valid service start slots across an extended horizon within persistent roster constraints.

Geographic Constraints Models start and end locations for every vehicle profile — optimising routes to end at a technician's home coordinate, reducing unpaid commute time and minimising depot-to-task mileage.

Non-Passing Property (FIFO) Enforces the FIFO constraint during duration matrix calculation, ensuring a vehicle departing later cannot arrive earlier than one that departed earlier — preventing "time-travel" errors and solver performance hangs in time-dependent environments.

Human-in-the-Loop: An Administrative Interface Layer sits above the vroom-express API, enabling dispatchers to review and override AI-generated skill classifications in edge cases — maintaining data integrity without sacrificing the speed of autonomous routing.

4 Business & Financial Impact Analysis

Optimising routing through this architecture directly impacts Operational Expenditure (OPEX) by maximising the efficiency of labour and fuel.

"Wrench Time" vs. "Windshield Time" Ratio

The system utilises the per_task_hour (setup + service time) and per_hour (travel time) cost keys within the VROOM vehicle object to prioritise active working hours over non-billable transit.

Increased Jobs per Day	Reduced Fuel Consumption	Lower Vehicle Wear
Maximising billable Wrench Time per technician per day via smarter task sequencing.	Optimising paths via dynamic speed_factor modifiers to avoid congested routes.	Minimising unproductive Windshield Time reduces mileage and maintenance overhead.

Cost Mitigation & Operational Efficiency

- **Cost Mitigation & SLA Compliance:** Running a local Dockerised solver avoids recurring per-query API costs. Enhanced ETA accuracy through temporal logic ensures higher SLA compliance, reducing late-arrival penalties.
- **Operational Efficiency:** Persistent rosters eliminate the "data cleanup" phase. Stateful job and engineer data bypasses manual verification steps required in ephemeral models, leading to faster daily deployment cycles.

5 Future Roadmap & Strategic Directions

- 1

Enhanced Multimodal Support (BETA)
Expanding Valhalla integration to include pedestrian and transit logic for urban last-mile deliveries where vehicle access is restricted or multimodal courier models are
- 2

Predictive Maintenance Integration
Utilising historical job data and the service_per_type API key to proactively schedule vehicle maintenance windows into the optimised route plan.
- 3

Real-Time Adaptive Rerouting
Implementing live response logic to TomTom traffic incidents, allowing the orchestrator to push updated vehicle_step arrays to drivers in real-time as road conditions change.

The VROOM platform has matured into an enterprise-grade dispatch leader. By synthesising AI-driven classification with persistent data architecture and FIFO-consistent temporal logic, it transforms logistics from a static cost centre into a high-performance, profit-generating engine.